

Simple IR Control RELAY SWITCH

RAJ K. GORKHALI

An IR remote switch is a wireless device that uses infrared (IR) signals from a remote control to operate electrical appliances. Commonly used in home automation, lighting control, and fan regulation, it enhances convenience and efficiency with seamless wireless switching.

The system consists of an IR receiver module, a signal processing circuit, and a relay to control the connected load. When an IR signal is detected, the receiver processes it and triggers a flip-flop circuit, activating the relay to switch the device on or off. Any standard IR remote control, including those for televisions, CD players, or MP3 players, can be used. Fig. 1 shows the author's prototype.

Circuit and working

Fig. 2 illustrates the circuit diagram of the IR remote switch. The circuit is built around the TSOP1738 IR module (IR1), IC 4027 (IC1), two PNP BC558 transistors (T1, T2), a 5V SPDT relay, and a few passive components. The 16-pin IC 4027, a dual JK flip-flop, serves as the core of the circuit, with only one section used for this application.

The circuit design is straightforward. Pin 2 of the TSOP1738 IR sensor connects to the positive terminal of the 5V power supply via a 100Ω resistor (R1), while capacitor C1 is placed between pin 2 and ground. Output pin 3 of the IR sensor is linked to the base of transistor T1 through resistor R2. The emitter of T1 is connected to the 5V power rail, and its collector is

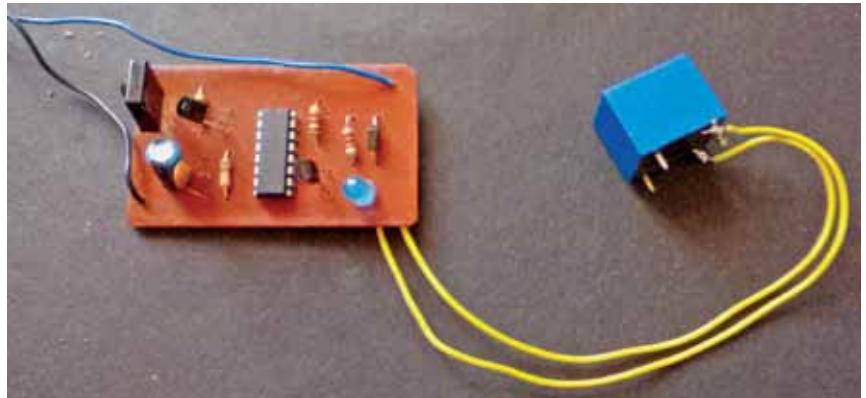


Fig. 1: Author's prototype

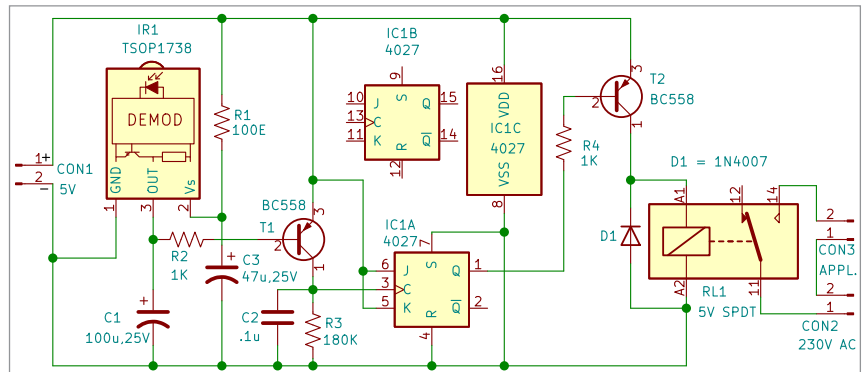


Fig. 2: Circuit diagram

grounded through resistor R3, with capacitor C2 connected in parallel. The collector of T1 also connects to pin 3 of IC1, serving as the clock input.

Pins 5, 6, and 16 of IC1 are connected to the 5V supply, while pins 4, 7, and 8 are grounded. The output from pin 1 of IC1 is fed to the base of transistor T2 via resistor R4. The emitter of T2 is connected to the 5V power rail, and its collector is linked to the ground through a 1N4001 rectifier diode. The cathode of the diode connects to the collector while the anode is grounded.

A 5V SPDT relay is placed between the collector of T2 and the ground.

When a button on the IR remote is pressed, the TSOP1738 sensor detects the signal, producing a low voltage at pin 3. This causes transistor T1 to conduct, sending a clock pulse to IC1, which generates a high output at pin 1. This output forward-biases transistor T2, energising the relay and turning the connected appliance on. Pressing the remote button again de-energises the relay, switching the appliance off.